Lab1: Transformations and Camera Projections (2h)

Computer Graphics: Rendering 2026

Question 1 — Translation and Homogeneous Coordinates

(a) Homogeneous coordinates

A 3D point can be represented using homogeneous coordinates as

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}.$$

The additional coordinate allows translation to be represented as matrix multiplication.

A general affine transformation can be written as

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} & & & t_x \\ & R & & t_y \\ & & & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix},$$

where R represents a linear transformation such as rotation or scaling.

Without the fourth coordinate, a 3×3 matrix can represent rotation and scale, but it cannot represent translation by simple matrix multiplication.

A direction vector is instead represented as

$$\mathbf{d} = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}.$$

The zero in the fourth coordinate means that translation does not affect the vector. For example,

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}.$$

Thus homogeneous coordinates distinguish between:

$$\boxed{\text{point} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}} \quad \text{and} \quad \boxed{\text{direction} = \begin{bmatrix} x \\ y \\ z \\ 0 \end{bmatrix}}.$$

(b) Translation matrix

We require

$$x' = x + t_x, \qquad y' = y + t_y, \qquad z' = z + t_z.$$

Let

$$T = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}.$$

We want

$$T \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ z + t_z \\ 1 \end{bmatrix}.$$

Therefore the first row must produce

$$ax + by + cz + d = x + t_x.$$

Hence

$$a = 1, \qquad b = 0, \qquad c = 0, \qquad d = t_x.$$

Similarly,

$$e = 0, \qquad f = 1, \qquad g = 0, \qquad h = t_y,$$

and

$$i = 0, \qquad j = 0, \qquad k = 1, \qquad l = t_z.$$

The final row must preserve the homogeneous coordinate, giving

$$m = n = o = 0, \qquad p = 1.$$

Thus

$$\boxed{T(t_x, t_y, t_z) = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

(c) Numerical calculation

We have

$$\mathbf{p} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

and

$$T = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore

$$T\mathbf{p} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$$

Multiplying gives

$$T\mathbf{p} = \begin{bmatrix} 1+4 \\ 2-2 \\ 3+5 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 8 \\ 1 \end{bmatrix}.$$

Hence

$$\boxed{T\mathbf{p} = \begin{bmatrix} 5 \\ 0 \\ 8 \\ 1 \end{bmatrix}.}$$

The corresponding 3D point is

$$\boxed{(5, 0, 8)}.$$

(d) Transformation order

The point is first rotated and then translated.

The 90° rotation around the z axis is

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The translation is

$$T = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since rotation occurs first, it appears on the right:

$$\boxed{M = TR_z(90^\circ).}$$

Multiplying,

$$M = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

so

$$\boxed{M = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

The order is important because matrix multiplication is not generally commutative:

$$TR \neq RT.$$

The rightmost matrix is applied first.

Question 2 — Rotation and Scale

(a) Rotation

We are given

$$x' = x\cos\theta - y\sin\theta,$$

$$y' = x\sin\theta + y\cos\theta,$$

and

$$z' = z.$$

The first equation tells us that the coefficients of x and y in the first row are $\cos\theta$ and $-\sin\theta$.
The second equation gives $\sin\theta$ and $\cos\theta$ in the second row.
The z coordinate is unchanged, so the third row is

$$[0 \quad 0 \quad 1].$$

Including the homogeneous coordinate gives

$$\boxed{R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.}$$

Indeed,

$$R_z(\theta)\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \\ z \\ 1 \end{bmatrix}.$$

(b) Scaling

We require

$$x' = s_x x, \qquad y' = s_y y, \qquad z' = s_z z.$$

Therefore

$$\boxed{S = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.}$$

For uniform scaling,

$$s_x = s_y = s_z = s,$$

so

$$\boxed{S = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & s & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.}$$

(c) Combined transformation

The point is first scaled and then rotated.

Therefore

$$\mathbf{p}' = R_z(\theta) S \mathbf{p}.$$

Hence

$$\boxed{M = R_z(\theta) S.}$$

The scaling matrix appears on the right because it is applied first.
Explicitly,

$$M = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore

$$\boxed{M = \begin{bmatrix} s_x \cos\theta & -s_y \sin\theta & 0 & 0 \\ s_x \sin\theta & s_y \cos\theta & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.}$$

(d) Numerical calculation

The original point is

$$\mathbf{p} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

After uniform scaling by 2,

$$\mathbf{p}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

For $\theta = 90^\circ$,

$$\cos 90^\circ = 0, \qquad \sin 90^\circ = 1,$$

so

$$R_z(90^\circ) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore

$$\mathbf{p}_2 = R_z(90^\circ)\mathbf{p}_1 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

Thus

$$\boxed{(x, y, z) = (0, 2, 0)}.$$

Question 3 — Orthographic Projection

(a) Derivation

We require

$$x' = sx, \qquad y' = sy, \qquad z' = z.$$

A suitable homogeneous matrix is

$$\boxed{P_{\text{ortho}} = \begin{bmatrix} s & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

Applying this matrix gives

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} sx \\ sy \\ z \\ 1 \end{bmatrix}.$$

The absence of any z term in the first two rows means that z does not affect x' or y'. In particular,

$$\frac{\partial x'}{\partial z} = 0, \qquad \frac{\partial y'}{\partial z} = 0.$$

(b) Projection of points

With

$$s = 2,$$

the projection equations are

$$x' = 2x, \qquad y' = 2y.$$

For

$$p_1 = (1, 2, -2),$$

we obtain

$$x'_1 = 2(1) = 2,$$

$$y'_1 = 2(2) = 4.$$

Thus

$$\boxed{p'_1 = (2, 4)}.$$

For

$$p_2 = (1, 2, -10),$$

we obtain

$$x'_2 = 2(1) = 2,$$

$$y_2' = 2(2) = 4.$$

Thus

$$\boxed{p_2' = (2, 4)}.$$

(c) Interpretation

Orthographic projection does not contain a depth-dependent division term.

The projected position is determined only by x and y:

$$(x, y, z) \longrightarrow (sx, sy).$$

Consequently, changing z while keeping x and y fixed does not change the projected position. This is why

$$(1, 2, -2)$$

and

$$(1, 2, -10)$$

project to the same image coordinate.

(d) Conceptual comparison

One advantage of orthographic projection is that objects do not change apparent size as their depth changes.

This makes it useful for applications where maintaining geometric proportions is important, such as:

- CAD;
- engineering drawings;
- technical visualisation;
- architectural drawings.

Question 4 — Perspective Projection

(a) Derivation

The camera is at the origin and looks along the negative z axis. Therefore a visible point has

$$z < 0.$$

Consider the point

$$P = (x, y, z)$$

and its projection

$$P' = (x', y', -f)$$

on a projection plane a distance f in front of the camera.
The two triangles formed by the camera centre and the point are similar. Therefore,

$$\frac{x'}{f} = \frac{x}{-z}.$$

Hence

$$\boxed{x' = f\frac{x}{-z}.}$$

The minus sign appears because the camera looks along negative z.

(b) Vertical coordinate

Applying exactly the same similar-triangle argument in the vertical direction gives

$$\frac{y'}{f} = \frac{y}{-z}.$$

Therefore

$$\boxed{y' = f\frac{y}{-z}.}$$

Thus the complete perspective projection is

$$\boxed{x' = f\frac{x}{-z}, \qquad y' = f\frac{y}{-z}.}$$

(c) Numerical calculation

Given

$$f = 2$$

and

$$p = (2, 1, -4),$$

we calculate

$$x' = 2\frac{2}{4} = 1.$$

Similarly,

$$y' = 2\frac{1}{4} = \frac{1}{2}.$$

Therefore

$$\boxed{p' = \left(1, \frac{1}{2}\right).}$$

(d) Depth and apparent size

For

$$p_1 = (2, 0, -2)$$

with $f = 1$,

$$x'_1 = \frac{2}{2} = 1.$$

For

$$p_2 = (2, 0, -8),$$

$$x'_2 = \frac{2}{8} = \frac{1}{4}.$$

Thus

$$\boxed{x'_1 = 1, \qquad x'_2 = \frac{1}{4}.}$$

The farther point has a larger magnitude of z. Consequently, the division by $-z$ produces a smaller projected coordinate.

This is the mathematical source of perspective foreshortening.

(e) Orthographic versus perspective

For orthographic projection,

$$\boxed{x' = sx, \qquad y' = sy.}$$

The projected coordinates do not depend on depth.

For perspective projection,

$$\boxed{x' = f\frac{x}{-z}, \qquad y' = f\frac{y}{-z}.}$$

The depth-dependent factor

$$\boxed{\frac{1}{-z}}$$

causes distant objects to appear smaller.

Question 5 — Constructing a Camera Ray from a Pixel

We use the following convention throughout:

$$\mathbf{C} = \text{camera position in world space},$$

$$R = \text{camera-to-world rotation},$$

and the camera looks along negative camera-space z.

(a) Pixel centre

Pixel (i, j) has its centre at

$$\left(i + \frac{1}{2}, j + \frac{1}{2}\right).$$

Therefore the normalized image coordinates are

$$\boxed{u = \frac{i + \frac{1}{2}}{W}, \qquad v = \frac{j + \frac{1}{2}}{H}.}$$

The half-pixel offset is required because (i, j) identifies the pixel cell, whose centre lies half a pixel from its top-left corner.

For example, the centre of pixel $(0, 0)$ is at

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

(b) Image plane

The image plane is at

$$z = -1.$$

The vertical field of view is θ. The optical axis intersects the image plane at its centre.

The half-height of the image plane is therefore

$$\frac{h}{2} = \tan\left(\frac{\theta}{2}\right).$$

Hence

$$\boxed{h = 2\tan\left(\frac{\theta}{2}\right).}$$

The aspect ratio is

$$\frac{W}{H}.$$

Therefore

$$\boxed{w = h\frac{W}{H}.}$$

Substituting for h,

$$\boxed{w = 2\tan\left(\frac{\theta}{2}\right)\frac{W}{H}.}$$

(c) Pixel to camera-space point

The image-plane horizontal coordinate runs from

$$-\frac{w}{2}$$

to

$$\frac{w}{2}.$$

Therefore normalized coordinate u maps to

$$x_c = \left(u - \frac{1}{2}\right) w.$$

For the vertical coordinate, $v = 0$ is at the top of the image while camera-space y increases upwards. Therefore

$$y_c = \left(\frac{1}{2} - v\right) h.$$

Hence

$$\boxed{x_c = \left(u - \frac{1}{2}\right) w}$$

and

$$\boxed{y_c = \left(\frac{1}{2} - v\right) h.}$$

The point on the image plane is therefore

$$\boxed{\mathbf{p}_c = \begin{bmatrix} \left(u - \frac{1}{2}\right) w \\ \left(\frac{1}{2} - v\right) h \\ -1 \end{bmatrix}.}$$

Substituting the pixel-coordinate expressions,

$$\boxed{x_c = \left(\frac{i + \frac{1}{2}}{W} - \frac{1}{2}\right) w}$$

and

$$\boxed{y_c = \left(\frac{1}{2} - \frac{j + \frac{1}{2}}{H}\right) h.}$$

(d) Camera-space ray

The ray originates at

$$\mathbf{o}_c = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore the unnormalized direction is

$$\mathbf{d}_c = \mathbf{p}_c - \mathbf{o}_c = \mathbf{p}_c.$$

Thus

$$\boxed{\mathbf{d}_c = \begin{bmatrix} x_c \\ y_c \\ -1 \end{bmatrix}.}$$

Its length is

$$\|\mathbf{d}_c\| = \sqrt{x_c^2 + y_c^2 + 1}.$$

The normalized direction is therefore

$$\boxed{\widehat{\mathbf{d}}_c = \frac{1}{\sqrt{x_c^2 + y_c^2 + 1}} \begin{bmatrix} x_c \\ y_c \\ -1 \end{bmatrix}.}$$

The camera-space ray is

$$\boxed{\mathbf{r}_c(t) = \mathbf{o}_c + t\widehat{\mathbf{d}}_c, \qquad t \geq 0.}$$

Since $\mathbf{o}_c = \mathbf{0}$,

$$\mathbf{r}_c(t) = t\widehat{\mathbf{d}}_c.$$

(e) Camera space to world space

For a point,

$$\mathbf{p}_w = R\mathbf{p}_c + \mathbf{C}.$$

A direction is different from a point: it represents a displacement, so translation must not affect it.

Therefore a camera-space direction transforms as

$$\boxed{\widehat{\mathbf{d}}_w = R\widehat{\mathbf{d}}_c.}$$

The camera-space origin of every camera ray is the camera itself. Thus

$$\boxed{\mathbf{o}_w = \mathbf{C}.}$$

Equivalently, since

$$\mathbf{o}_c = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

we have

$$\mathbf{o}_w = R\mathbf{o}_c + \mathbf{C} = \mathbf{C}.$$

Hence the world-space ray is

$$\boxed{\mathbf{r}_w(t) = \mathbf{C} + tR\widehat{\mathbf{d}}_c, \qquad t \geq 0.}$$

Translation is not added to the direction because a direction is not a position.

(f) Numerical example

We are given

$$\mathbf{C} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \qquad R = I,$$

$$\theta = 90^\circ, \qquad W = 400, \qquad H = 400,$$

and

$$(i, j) = (199, 199).$$

Step 1: Pixel centre We have

$$u = \frac{199 + \frac{1}{2}}{400} = \frac{199.5}{400} = 0.49875,$$

and

$$v = \frac{199 + \frac{1}{2}}{400} = 0.49875.$$

Thus

$$\boxed{u = v = 0.49875.}$$

Step 2: Image-plane dimensions Since

$$\theta = 90^\circ,$$

we have

$$h = 2 \tan\left(\frac{90^\circ}{2}\right) = 2 \tan 45^\circ = 2.$$

Since the image is square,

$$w = h\frac{W}{H} = 2\frac{400}{400} = 2.$$

Therefore

$$\boxed{w = h = 2.}$$

Step 3: Camera-space image-plane point The horizontal coordinate is

$$x_c = \left(u - \frac{1}{2}\right) w.$$

Therefore

$$x_c = (0.49875 - 0.5)(2) = -0.0025.$$

Similarly,

$$y_c = \left(\frac{1}{2} - v\right) h$$

gives

$$y_c = (0.5 - 0.49875)(2) = 0.0025.$$

Thus

$$\boxed{\mathbf{p}_c = \begin{bmatrix} -0.0025 \\ 0.0025 \\ -1 \end{bmatrix}.}$$

Step 4: Camera-space ray direction The unnormalized direction is

$$\mathbf{d}_c = \begin{bmatrix} -0.0025 \\ 0.0025 \\ -1 \end{bmatrix}.$$

Its magnitude is

$$\|\mathbf{d}_c\| = \sqrt{(-0.0025)^2 + (0.0025)^2 + (-1)^2}.$$

Therefore

$$\|\mathbf{d}_c\| = \sqrt{1.0000125} \approx 1.00000625.$$

Hence

$$\widehat{\mathbf{d}}_c = \frac{1}{1.00000625} \begin{bmatrix} -0.0025 \\ 0.0025 \\ -1 \end{bmatrix}.$$

Numerically,

$$\boxed{\widehat{\mathbf{d}}_c \approx \begin{bmatrix} -0.00249998 \\ 0.00249998 \\ -0.99999375 \end{bmatrix}.}$$

Step 5: Transform to world space Since

$$R = I,$$

the direction is unchanged:

$$\widehat{\mathbf{d}}_w = R\widehat{\mathbf{d}}_c = \widehat{\mathbf{d}}_c.$$

The world-space origin is

$$\mathbf{o}_w = \mathbf{C} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

Therefore

$$\boxed{\widehat{\mathbf{d}}_w \approx \begin{bmatrix} -0.00249998 \\ 0.00249998 \\ -0.99999375 \end{bmatrix}.}$$

The final world-space ray is

$$\boxed{\mathbf{r}_w(t) = \begin{bmatrix}2\\3\\5\end{bmatrix} + t\begin{bmatrix}-0.00249998\\0.00249998\\-0.99999375\end{bmatrix}, \qquad t \geq 0.}$$

Equivalently,

$$\boxed{\begin{aligned}x(t) &= 2 - 0.00249998t,\\ y(t) &= 3 + 0.00249998t,\\ z(t) &= 5 - 0.99999375t.\end{aligned}}$$

The ray is almost parallel to the negative world-space z axis, because the selected pixel is very close to the centre of the image.